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SUBJECT OF THE INVENTION

The radioactive emanation collecting device [APPARATUS FOR CAPTURING NUCLEAR GAS IN PRESSURE VESSEL OF REACTOR] remaining behind within the pressure vessel of the system of atomic pile.

[**0001**] The present invention relates to the radioactive emanation collecting device remaining behind within the pressure vessel of the system of atomic pile, and it relates to the radioactive emanation collecting device removing solubility, and the insoluble radioactive emanation and emits the gas without radioactivity into the radioactive emanation remaining behind within the pressure vessel to the outside.

[**0002**] The atomic power plant stops the nuclear reactor among the organizing period and the major pressure vessel of the system of atomic pile including the nuclear reactor, the steam generator, the pressurizer etc. is opened for the fuel exchange, the flushing and inspection. Even in this case, the radioactive emanation can remain behind inside the pressure vessel. In this state, when the pressure vessel is opened until the polluted air is this cleaned up while the housing building inside is polluted due to the air the air need the long time. Moreover, it has the problem the access of the operator is controlled until the air of the housing building inside is cleaned up and accordingly that the organizing period grows longer.

[**0003**] Nevertheless, before the current atomic power plant opens the major pressure vessel including the container, the steam generator, the pressurizer etc. to the atom. The apparatus, for collecting the radioactive emanation remaining behind inside the pressure vessel and removing and the step of performing the same are the a bit insufficient actual condition.

[**0004**] The content removing halogen to the scrubber processing and waste water is written the gas in which JP5320323 B (" waste water processing method and processing unit ", 2013.07.19 B, and the prior art) are generated during the atomic power facility in connection with this. However, the prior art has solubility and the problem that the content removing the insoluble radioactive emanation and ejects the pure gas is not printed and it does not feed back to the again when radioactivity is measured in the ejected gas and the radioactive emanation is exhausted.

[**0005**] JP5320323 B ("the waste water processing method and processing unit", and 2013.07.19 B)

[**0006**] The invention is provided to solve the above-described problem. And an object of the present invention is to provide the radioactive emanation collecting device remaining behind with the system of atomic pile pressure vessel inside for the effectively collecting the radioactive emanation among the gas component remaining behind including the nuclear reactor inside the major pressure vessel of the system of atomic pile and exhausting after the purification.

[**0007**] Moreover, the invention is to provide the radioactive emanation collecting device which measures the radioactivity whether or not before exhausting the gas from which the radioactive emanation is removed and when radioactivity is measured remains behind. For feeding back to the again and completely removing the radioactive emanation.

[**0008**] The present invention is comprised of the exhausting unit (400) which it is connected by the pressure vessel (100): first connector (110) in which the radioactive emanation is received within the pressure vessel of the system of atomic pile as the radioactive emanation collecting device remaining behind to the pressure vessel (100) and the radioactive emanation is flowed in from the pressure vessel (100) and it is connected by the scrubber (200): second connection part (210) removing the solubility radioactive emanation among the radioactive emanation flowed in to the scrubber (200) and the radioactive emanation is flowed in from the scrubber (200) and it is connected by the filter section (300): third connection part (310) removing the insoluble radioactive emanation from the scrubber (200) among the radioactive emanation flowed in to the filter section (300) and solubility and the gas from which the insoluble radioactive emanation is removed are flowed in from the filter section (300) and it ejects the gas according to the radioactivity whether or not of the gas flowed in from the filter section (300) to the outside or it blocks.

[**0009**] Moreover, the scrubber (200) comprises the solution (201) which is received inside the scrubber (200) and it reacts with the radioactive emanation flowed in from the pressure vessel (100) and removes the solubility radioactive emanation; the first radiometric analyzer (202) measuring the radioactivity of the solution (201); the solution level measuring device (204) measuring the water level of the first thermoscope (203): solution (201) measuring the temperature of the solution (201).

[**0010**] Moreover, it further includes the temperature controller (205) in which the scrubber (200) regularly controls the temperature of the solution (201).

[**0011**] Moreover, the filter section (300) comprises the filter (301) which is equipped inside the filter section (300) and collects the insoluble radioactive emanation among the radioactive emanation

flowed in from the scrubber (200); the second radiation line measuring instrument (302) measuring the radioactivity of the filter section (300); the second thermoscope (303): cooling part (304) cooling the filter section (300) measures the temperature of the filter section (300).

[**0012**] Moreover, the exhausting unit (400) may be provided in the flow meter (404): delay coil (401) which is equipped in the front end of the delay coil (401): delay coil (401) which is formed with the spring shape and delays that the gas flowed is exhausted from the filter section (300) to the outside and it is equipped in the rear end of the third radiometric analyzer (402): delay coil (401) measuring the radioactivity of the gas flowed in from the filter section (300) and it is equipped in the shutoff valve (403): delay coil (401) ejecting the gas according to the radiation measurement value which measures in the third radiometric analyzer (402) to the outside or blocked and measures the flow rate of the gas. And comprises the third thermoscope (405) measuring the temperature of the gas.

[**0013**] Moreover, it further includes the feedback line (410) which when the exhausting unit (400) the radioactivity of the gas is measured is connected to the first connector (110) in order to move the gas to the first connector (110).

[**0014**] Moreover, it further includes the gas flowing backward prevention means (411) in which the feedback line (410) prevents that the gas flows backward.

[**0015**] Moreover, the first connector (110) comprises the pressure measuring device (111) measuring the pressure of the first connector (110); the flow rate controller (112) which regularly controls the flow rate of the radioactive emanation flowed in from the pressure vessel (100); the fourth thermoscope (114) measuring the temperature of the fourth radiometric analyzer (113): radioactive emanation flowed in from the pressure vessel (100) measures the radioactivity of the radioactive emanation flowed in from the pressure vessel (100).

[**0016**] Moreover, it further includes the gas injector (115) in which the first connector (110) injects the gas into the pressure vessel (100).

[**0017**] Moreover, the second pressure meter (211) in which the second connection part (210) measures the pressure of the second connection part (210) is equipped.

[**0018**] Moreover, the third pressure measuring device (311) in which the third connection part (310) measures the pressure of the third connection part (310) is equipped.

[**0019**] Moreover, the solution (201) is the alkaline solution more than the pH 10.

[**0020**] Moreover, the filter (301) comprises the activated charcoal or the MOF (metal-organic framework) material.

[**0021**] The present invention relates to the radioactive emanation collecting device remaining behind with the system of atomic pile pressure vessel inside and has the effect that the radioactive emanation remaining in the system of atomic pile major pressure vessel by using the sewage disposal facility of two stages is collected and the removal is facilitated. Especially, it has the effect that collects the radioactive emanation in which the radiodine, xenon, and the krypton lamp diffusion mobility are high and removed.

[**0022**] Moreover, the present invention has the effect that the safety of the working employee is secured among the nuclear power plant plan and surprising organizing period from the radioactive emanation and the extension of the nuclear power plant organizing period generated in the radioactive emanation contamination can be beforehand prevented and it contributes to the safety assurance of the nuclear power plant main part side since the radioactive emanation is removed to the system of atomic pile outside without the leakage.

[**0023**] Moreover, the present invention has the effect that the radioactivity whether or not of the gas is confirmed before being exhausted and when radioactivity is detected it feeds back to the again and the radioactive emanation is completely removed.

[**0024**] Figure 1 is a schematic diagram of the apparatus for collecting the radioactive emanation remaining behind within the pressure vessel of the system of atomic pile according to the invention

[**0025**] Hereinafter, more specifically, the technical mapping of the invention is illustrated using the accompanying drawing.

[**0026**] Since it is nothing but one example shown in order to specifically illustrate the accompanying drawing is the technical mapping of the invention, the technical mapping of the invention is not restricted to the form of the accompanying drawing.

[**0027**] Figure 1 is a schematic diagram of the apparatus for collecting the radioactive emanation remaining behind within the pressure vessel of the system of atomic pile according to the invention.

[**0028**] Referring to Figure 1, it illustrates. And the present invention relates to the apparatus for collecting the radioactive emanation remaining behind within the pressure vessel of the system of atomic pile, and the scrubber (200) is connected to the pressure vessel (100) in which the radioactive emanation remains behind and the filter section (300) is connected to the scrubber (200) and the exhausting unit (400) is connected to the filter section (300). More specifically, the pressure vessel (100) and the scrubber (200) in which the radioactive emanation remains behind are connected by the first connector (110) and the radioactive emanation of the pressure vessel (100) inside is flowed into the scrubber (200). Then, the solubility radioactive emanation is removed among the radioactive emanation in which the scrubber (200) is flowed in.

[**0029**] The filter section (300) is connected by the scrubber (200) and the second connection part (210) and the radioactive emanation from which the solubility radioactive emanation is removed is flowed in. Then, the insoluble radioactive emanation is collected among the radioactive emanation in which the filter section (300) is flowed in.

[**0030**] The exhausting unit (400) is connected by the filter section (300) and the third connection part (310) and the gas from which the insoluble radioactive emanation is removed is flowed in. Then, according to the radioactivity whether or not of the gas in which the exhausting unit (400) is flowed in, the gas is ejected or it blocks.

[**0031**] The pressure vessel (100) refers to the major pressure vessel of the system of atomic pile including the nuclear reactor, the steam generator, the pressurizer etc. If the nuclear reactor stops , the volatile radioactive emanation remains behind inside the pressure vessel (100). The iodine (I_2) gas in which the radioactive emanation is the solubility radioactive emanation and the Xe, which is the insoluble radioactive emanation the Kr, the organic iodine etc. exist.

[**0032**] The scrubber (200) is connected by the first connector (110) to the pressure vessel (100) and the solubility radioactive emanation serves to be removed from the pressure vessel (100) among the radioactive emanation flowed in.

[**0033**] The scrubber (200) the solution (201) in which the solubility radioactive emanation is dissolved is adopted to the inside. It is the aqueous solution in which the main component of the solution (201) is made of the reducing agent ($Na_2S_2O_3$), and the alkali material (NaOH, Na_3PO_4). The alkali material (NaOH, Na_3PO_4) is the pH 10 or greater. The iodide which is stable the iodine (I_2) gas which is the solubility gas among the radioactive emanation in which the solution (201) is flowed in

through the chemical reaction is diversified and it is dissolved in the solution (201). Due to this, the radioactive emanation reacting to the solution (201) removes the solubility radioactive emanation.

[**0034**] Moreover, the scrubber (200) the first radiometric analyzer (202), the first thermoscope (203), and the solution level measuring device (204) and temperature controller (205) are equipped. The first radiometric analyzer (202) measures the radioactivity of the solution (201) and is used to bring forth amount of the sampled radioactive emanation within the solution (201). Moreover, the first thermoscope (203) measures the temperature of the solution (201) and the temperature controller (205) controls so that the temperature of the solution (201) is maintained by 20 ~ 60 ℃. Moreover, the solution level measuring device (204) measures the water level of the solution (201) and the occurrence of the given water level or greater or the solution (201) is perceived. Moreover, in the scrubber (200) , the separator (not illustrated) is further provided at the inside. And moisture and gas can be separated using the cyclone phenomenon.

[**0035**] The filter section (300) is connected by the second connection part (210) to the scrubber (200) and the insoluble radioactive emanation serves to be collected in the scrubber (200) among the radioactive emanation flowed in. The filter section (300) the filter (301), the second radiation line measuring instrument (302), and the second thermoscope (303) and cooling part (304) are equipped.

[**0036**] The filter (301) is equipped inside the filter section (300) and the insoluble radioactive emanation serves to be collected in the scrubber (200) among the radioactive emanation flowed in. Moreover, the material of the filter (301) comprises the activated charcoal or the MOF (metal-organic framework).

[**0037**] The second radiation line measuring instrument (302) measures the radioactivity of the filter section (300) and is used to bring forth amount of the radioactive emanation sampled in the filter (301).

[**0038**] The cooling part (304) serves to cool the filter section (300) and it is equipped in the outside of the filter section (300). The adhesion is degraded as the temperature the filter (301) rises. The cooling part (304) maintains the temperature of the filter section (300) less than 60℃ and the adhesion of the filter (301) can be secured.

[**0039**] The exhausting unit (400) is connected by the third connection part (310) to the filter section (300) and solubility and the gas from which the insoluble radioactive emanation is removed serve to be

ejected to the outside. The exhausting unit (400) the delay coil (401), third radiometric analyzer (402), shutoff valve (403), flow meter (404) and the third thermoscope (405) are equipped.

[**0040**] When ejecting the gas in which the delay coil (401) is flowed in from the filter section (300) to the outside, the drain time serves to be delayed. For this, it is desirable that the length is formed longer than like the zigzag or the spring shape in which the delay coil (401) is not straight geometry.

[**0041**] The radioactivity of the gas in which the third radiometric analyzer (402) is flowed into the filter section (300) is measured and the gas is emitted according to the measured value of the radioactivity measured in the third radiometric analyzer (402) to the outside or the shutoff valve (403) blocks. That is, in the third radiometric analyzer (402), when radioactivity is measured it blocks so that the shutoff valve (403) is unable to be the gas ejected to the outside. On the other hand, when radioactivity is not measured the shutoff valve (403) is opened so that the gas is exhausted to the outside.

[**0042**] The flow meter (404) measures the flow rate of the gas and the thermoscope measures the temperature of the gas. Amount of moisture can be produced among the temperature measured in the flow meter (404) and thermoscope and the gas in which SCCM is exhausted and the evaporation degree of the solution (201) can be predicted in the scrubber (200).

[**0043**] Moreover, the feedback line (410) in which the exhausting unit (400) connects the rear end and the first connector (110) of the exhausting unit (400) is equipped. In the feedback line (410) is the gas, when radioactivity is measured the gas serves to be again moved to the first connector (110). More specifically, in the exhausting unit (400), if the radioactivity of the gas is measured , the shutoff valve (403) is closed and the gas is moved to the first connector (110) through the feedback line (410). Moreover, it further includes the feedback line (410) is the gas flowing backward prevention means (411). In the gas flowing backward prevention means (411) is the first connector (110), the radioactive emanation does not move the shutoff valve (403) and the check valve or the pump etc. can be used.

[**0044**] The first connector (110) is positioned between the pressure vessel (100) and scrubber (200) and the pressure vessel (100) and scrubber (200) are connected. One end the first connector (110) is combined with one side of the pressure vessel (100) and the other end is formed in order to be sunk under the solution (201) within the scrubber (200). Moreover, the first connector (110) the pressure measuring device (111), the flow rate controller (112), and the fourth radiometric analyzer (113) and fourth thermoscope (114) are equipped.

[**0045**] The pressure measuring device (111) measures the pressure of the first connector (110) and the pressure of the pressure vessel (100) can be produced. The flow rate of the gas in which the flow rate controller (112) is flowed in serves to be controlled. That is, the flow rate controller (112) the flow rate of the gas regularly controls regardless of the pressure of the pressure vessel (100). In the fourth radiometric analyzer (113), the radioactive emanation the pressure vessel (100), when it moves to the scrubber (200) , the radioactivity of the radioactive emanation is measured. Moreover, the fourth thermoscope (114) measures the temperature of the radioactive emanation.

[**0046**] Moreover, it further includes the fastening member (not illustrated) in which the first connector (110) is connected to the pressure vessel (100). The fastening member can use the clamp or the fixing screw etc. When the radioactive emanation of the pressure vessel (100) inside such fastening member is moved to the scrubber (200) , it prevents the radioactive emanation from being leaked out to the outside.

[**0047**] Moreover, it further includes the first connector (110) is the gas injector (115). The gas injector (115) injects the gas into the pressure vessel (100) and the pressure serves to be increased. More specifically, when the pressure of the pressure vessel (100) inside is lower than that of the pressure of the scrubber (200) , the gas can move from the scrubber (200) the pressure vessel (100). To prevent this problem, the gas injector (115) injects the gas into the pressure vessel (100) and the pressure of the pressure vessel (100) inside is increased and the radioactive emanation of the pressure vessel (100) inside is moved the scrubber (200). Moreover, it is desirable to use the gas in which the gas of the gas injector (115) does not react to the radioactive emanation, the solution (201), and the filter (301) and the inert gas like the argon (Ar) and nitrogen (N) can be used.

[**0048**] The second connection part (210) connects the scrubber (200) and filter section (300) and the second pressure meter (211) is equipped. The second pressure meter (211) serves to measure the pressure of the second connection part (210) and the measured value measured in the second pressure meter (211) and the measured value measured in the pressure measuring device (111) are compared and the occurrence and normal operation whether or not of the scrubber (200) inside solution (201) can be grasped by the pressure difference.

[**0049**] The third connection part (310) connects the filter section (300) and exhausting unit (400) and the third pressure measuring device (311) is equipped. The third pressure measuring device (311) serves to measure the pressure of the third connection part (310) and the measured value measured in the third pressure measuring device (311) and the measured value measured in the

second pressure meter (211) are compared and the performance of the filter (301) can be grasped by the pressure difference.

[**0050**] The invention is not restricted in the above-described embodiment and without deviating from the gist of the invention claimed in the claims the applicability is varied capable of the various transformation execution is of course.

[**0051**] 100: pressure vessel.
110: first connector.
111: pressure measuring device.
112: flow rate controller.
113: fourth radiometric analyzer.
114: fourth thermoscope.
115: gas injector.
200: scrubber.
201: solution.
202: first radiometric analyzer.
203: first thermoscope.
204: solution level measuring device.
205: temperature controller.
210: second connection part.
211: second pressure meter.
300: filter section.
301: filter.
302: second radiation line measuring instrument.
303: second thermoscope.
304: cooling part.
310: third connection part.
311: third pressure measuring device.
400: exhausting unit.
401: delay coil.
402: third radiometric analyzer.
403: shutoff valve.
404: flow meter.
405: third thermoscope.
410: feedback line.
411: gas flowing backward prevention means.

It is connected by the pressure vessel (100): first connector (110) in which the radioactive emanation is received to the pressure vessel (100); the radioactive emanation is flowed in from the pressure vessel (100); it is connected by the scrubber (200): second connection part (210) removing the solubility radioactive emanation among the radioactive emanation flowed in to the scrubber (200); the radioactive emanation is flowed in from the scrubber (200); it is connected by the filter section (300): third connection part (310) removing the insoluble radioactive emanation from the scrubber (200) among the radioactive emanation flowed in to the filter section (300); and solubility and the

gas from which the insoluble radioactive emanation is removed are flowed in from the filter section (300). The exhausting unit (400) the exhausting unit (400) which it ejects the gas according to the radioactivity whether or not of the gas flowed in from the filter section (300) to the outside is included may be provided in the flow meter (404): delay coil (401) which is equipped in the front end of the delay coil (401): delay coil (401) which is formed with the spring shape; and delays that the gas flowed is exhausted from the filter section (300) to the outside; it is equipped in the rear end of the third radiometric analyzer (402): delay coil (401) measuring the radioactivity of the gas flowed in from the filter section (300); it is equipped in the shutoff valve (403): delay coil (401) ejecting the gas according to the radiation measurement value which measures in the third radiometric analyzer (402) to the outside or blocked; and measures the flow rate of the gas. And the radioactive emanation collecting device remaining behind within the pressure vessel of the system of atomic pile which comprises the third thermoscope (405) measuring the temperature of the gas.

As for claim 1, the radioactive emanation collecting device remaining behind within the pressure vessel of the system of atomic pile which comprises the first radiometric analyzer (202), the first thermoscope (203) measuring the temperature of the solution (201), and the solution level measuring device (204) measuring the water level of the solution (201) wherein the scrubber (200) measures the radioactivity of the solution (201): solution (201) that removes the solubility radioactive emanation it reacts with the radioactive emanation flowed in from the pressure vessel (100) it is received inside the scrubber (200).

As for claim 2, the radioactive emanation collecting device remaining behind within the pressure vessel of the system of atomic pile in which the temperature controller (205) in which the scrubber (200) regularly controls the temperature of the solution (201) is further equipped.

As for claim 1, the radioactive emanation collecting device remaining behind within the pressure vessel of the system of atomic pile which comprises the second radiation line measuring instrument (302), the second thermoscope (303) measuring the temperature of the filter section (300), and the cooling part (304) cooling the filter section (300) wherein the filter section (300) measures the radioactivity of the filter (301): filter section (300) that collects the insoluble radioactive emanation among the radioactive emanation which is equipped inside the filter section (300); and is flowed in from the scrubber (200).

Deletion.

As for claim 1, the radioactive emanation collecting device which the exhausting unit (400) the radioactivity of the gas is measured; and remains behind within the pressure vessel of the system of atomic pile in which the feedback line (410) which is connected to the first connector (110) in order to move the gas to the first connector (110), is further equipped.

As for claim 6, the radioactive emanation collecting device remaining behind within the pressure vessel of the system of atomic pile in which the gas flowing backward prevention means (411) in which the feedback line (410) prevents that the gas flows backward is further equipped.

As for claim 1, the radioactive emanation collecting device remaining behind within the pressure vessel of the system of atomic pile which comprises the flow rate controller (112), the fourth radiometric analyzer (113) measuring the radioactivity of the radioactive emanation flowed in from the pressure vessel (100), and the fourth thermoscope (114) measuring the temperature of the radioactive emanation flowed in from the pressure vessel (100) wherein the first connector (110) regularly controls the flow rate of the radioactive emanation flowed in from the pressure measuring device (111): pressure vessel (100) that measures the pressure of the first connector (110).

As for claim 8, the radioactive emanation collecting device in which the first connector (110) remains behind within the pressure vessel of the system of atomic pile in which the gas injector (115) injecting the gas into the pressure vessel (100) is further equipped.

As for claim 1, the radioactive emanation collecting device in which the second connection part (210) remains behind within the pressure vessel of the system of atomic pile in which the second pressure meter (211) measuring the pressure of the second connection part (210) is equipped.

As for claim 1, the radioactive emanation collecting device in which the third connection part (310) remains behind within the pressure vessel of the system of atomic pile in which the third pressure measuring device (311) measuring the pressure of the third connection part (310) is equipped.

As for claim 2, solution (201) The radioactive emanation collecting device remaining behind within the pressure vessel of the system of atomic pile which is the alkaline solution more than the pH 10.

As for claim 4, the radioactive emanation collecting device remaining behind within the pressure vessel of the system of atomic pile in which the filter (301) is made of the activated charcoal or the MOF material.